

Final Report: Paper & Project

Student Name: XXX

Paper: “Can Local Regularization Learn All Multiclass Problems?” by Asilis, Devic, Dughmi, Sharan, and Teng. Published at COLT 2024 (PMLR Vol. 196, pp. 1–5).

Project: Python experimental framework that implements and tests the paper’s constructions.

1 What This Paper Is About

This paper looks at a pretty basic question: is there a simple learning rule for multiclass classification, the way ERM (empirical risk minimization) works for binary classification?

In binary classification, things are clean. You have two labels, VC dimension tells you if a class is learnable, and ERM — just picking any hypothesis that gets zero training error — works great. This has been known since the late 1980s.

Multiclass is where things get messy. When you have more than two labels (maybe even infinitely many), ERM breaks down. Daniely and Shalev-Shwartz showed in 2014 that some learnable multiclass classes *require* the learner to output hypotheses outside the hypothesis class $\mathcal{H}$. That means ERM, which only picks from $\mathcal{H}$, is fundamentally limited.

So the paper asks: can **local regularization** fill this gap? Local regularization is a variant of structural risk minimization (SRM) where the regularizer $\psi(h, x)$ depends on both the hypothesis h and the test point x . This extra flexibility lets the learner be “improper” — it can stitch together predictions from different hypotheses at different points.

2 Background Concepts

2.1 PAC Learning

The standard setup: we have a domain $\mathcal{X}$, labels $\mathcal{Y}$, and a hypothesis class $\mathcal{H} \subseteq \mathcal{Y}^{\mathcal{X}}$. A learner gets a training sample $S = \{(x_1, y_1), \dots, (x_n, y_n)\}$ drawn i.i.d. from some distribution D , and needs to find a hypothesis with low true error $L_D(h) = \Pr_{(x,y) \sim D}[h(x) \neq y]$.

A learner is a PAC learner if there’s a sample size $m(\epsilon, \delta)$ such that with enough data, it achieves error $\leq \epsilon$ with probability $\geq 1 - \delta$.

2.2 Why ERM Fails for Multiclass

ERM is a *proper* learner — it only outputs $h \in \mathcal{H}$. SRM adds a regularizer $\psi(h)$ but is still proper. The problem is that some multiclass classes need *improper* learners that can output predictions not matching any single hypothesis.

The current best learners use one-inclusion graphs (OIGs), which are mathematically powerful but completely impractical — they can be exponentially large or even infinite.

2.3 Local Regularization

Definition 1 A *local regularizer* for $\mathcal{H}$ is a function $\psi : \mathcal{H} \times \mathcal{X} \rightarrow \mathbb{R}_{\geq 0}$. It induces a learner that, given sample S and test point x , outputs:

$$\mathcal{A}(S)(x) \in \left\{ h(x) : h \in \arg \min_{h \in L_S^{-1}(0)} \psi(h, x) \right\}$$

where $L_S^{-1}(0)$ is the set of hypotheses with zero training error.

The key difference from SRM: ψ depends on x , not just h . This means:

- It can say “ h_0 is simple here but complex there”
- Different hypotheses can “win” at different points, making the learner improper
- It doesn’t need unlabeled training data (unlike OIGs)

One important note: there’s also an *unsupervised* version of local regularization where ψ can also look at the unlabeled x -values in the training set. That version was already shown to learn everything. The open question is about the plain version that can’t peek at the training data.

3 The Candidate Counterexample: $\mathcal{H}_\Delta$

To try to show that local regularization is NOT enough, the authors construct a specific hypothesis class $\mathcal{H}_\Delta$.

3.1 How It’s Built

- Domain: $\mathcal{X} = \mathbb{N}$ (natural numbers)
- Labels: $\mathcal{Y} = \{*\} \cup 2^{\mathcal{X}}$ (a default “star” label plus subsets of $\mathcal{X}$)
- Take triples of sets (A, B, C) where:
 - $|A| = |B| = |C| = k$ (all same size)
 - $|A \cap B| = |A \cap C| = |B \cap C| = k/2$ (pairwise overlaps are half)
 - $A \cap B \cap C = \emptyset$ (no triple overlap)
- Each triple gives 8 hypotheses — one for each way to assign labels on the three pairwise intersections ($2 \text{ choices} \times 3 \text{ regions} = 2^3 = 8$)
- Outside $A \cup B \cup C$, every hypothesis just predicts $*$

The intuition: think of three sheets of paper labeled A, B, C laid on top of each other. A hypothesis is a choice of which sheet goes on top in each overlap region. Looking down from above, you see the top sheet’s name.

3.2 It IS Learnable

$\mathcal{H}_\Delta$ can be learned by a simple “Default-Star” learner:

1. Default: predict * everywhere
2. If you’ve seen a label A containing x in the training data, predict A
3. If you’ve seen two labels both containing x , use the training data to figure out which is correct

This works because once you’ve seen a data point from each overlap region, you know the correct layering.

3.3 Why ERM Fails On It

Say the true distribution is uniform over $\{(x, A) : x \in A\}$ for some large set A . If the training set has $< |A|/2$ points, with high probability a test point x_{test} was not seen. For that unseen point, you can find a triple (A, B, C) where $x_{\text{test}} \in B$ and B avoids the training set. So there’s a zero-error hypothesis predicting $B \neq A$ at x_{test} . ERM might pick it, and it’s wrong.

3.4 Why Local Reg Might Also Fail

A local regularizer $\psi(h, x)$ only knows h and x_{test} . It can NOT look at the training sample. When two zero-error hypotheses exist — one predicting A (correct) and one predicting B (wrong) — the regularizer sees the same information for both. The symmetry of $\mathcal{H}_\Delta$ prevents it from telling them apart.

4 The Open Problems

Open Problem 1: Can all learnable multiclass hypothesis classes be learned by a local regularizer? If yes, with near-optimal sample complexity?

Open Problem 2: Can $\mathcal{H}_\Delta$ specifically be learned by a local regularizer?

If you show $\mathcal{H}_\Delta$ is NOT learnable by any local regularizer, that settles Problem 1 negatively too.

5 My Project: Python Implementation

For the project, I built a Python framework to experimentally test the paper’s claims.

5.1 What I Implemented

Four main files:

- `hypothesis_class.py` — Builds $\mathcal{H}_\Delta$: generates valid triples, creates 8 hypotheses per triple, handles sampling
- `learners.py` — Three learner types: ERM (picks random zero-error hypothesis), Default-Star (the correct learner), and Local Regularizer (with different ψ strategies)
- `pac_framework.py` — Runs PAC evaluation: draws samples, computes errors over many trials
- `run_experiments.py` — Runs 5 experiments and makes plots

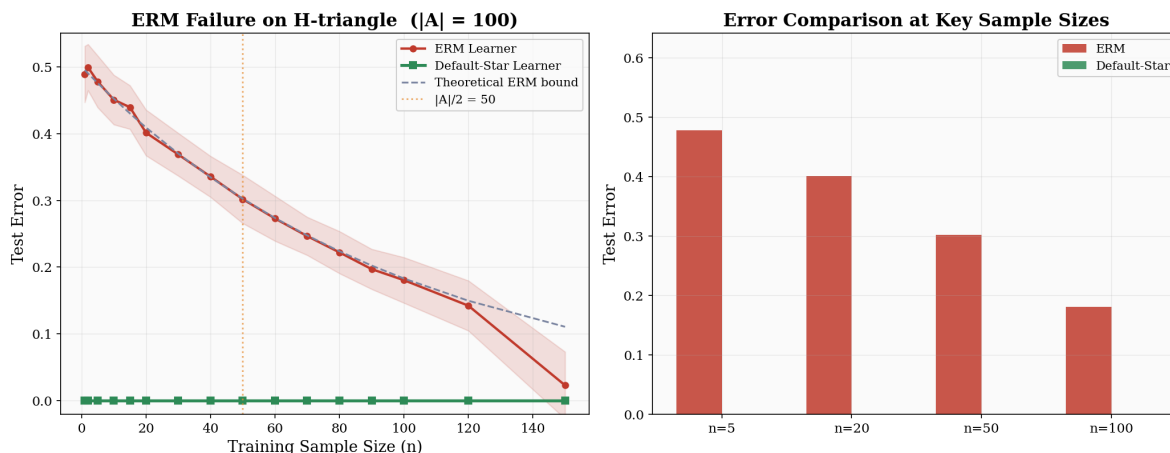
5.2 How the Simulation Works

Instead of building a huge finite $\mathcal{H}_\Delta$ and hoping ERM stumbles onto adversarial hypotheses, I directly simulate the paper’s argument:

1. Fix $A = \{0, 1, \dots, k-1\}$, distribution = uniform over $\{(x, A) : x \in A\}$
2. Draw n training points, track which elements are “seen”
3. For each test point: if unseen, an adversarial hypothesis exists, ERM/local-reg picks it wrong with probability $\sim 1/2$
4. Default-Star always gets it right (knows A from any single training point)

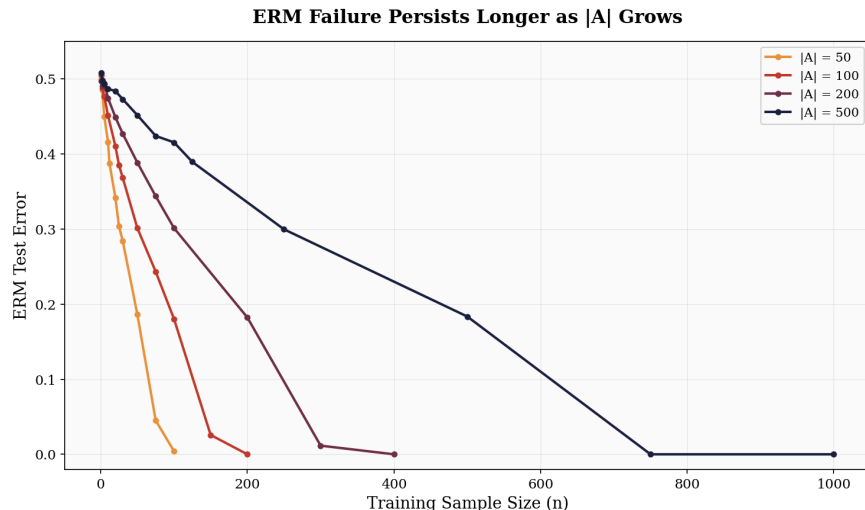
6 Experimental Results

6.1 Experiment 1: ERM Failure



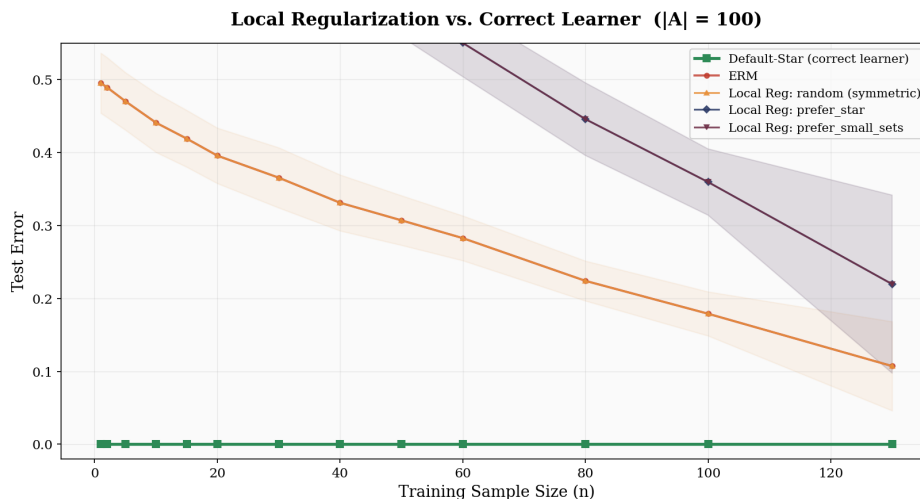
ERM (red) starts near 50% error and slowly declines. The dashed line is the theoretical bound $(1 - 1/k)^n \times 0.5$, which matches closely. Default-Star (green) is at zero from the start. The bar chart on the right shows the gap at specific sample sizes — even at $n = 100$, ERM has $\sim 18\%$ error.

6.2 Experiment 2: Scaling



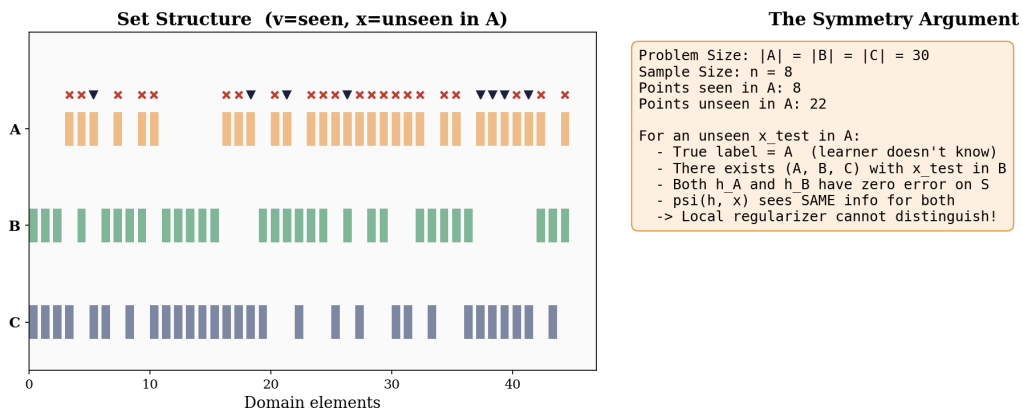
ERM error for $|A| = 50, 100, 200, 500$. Bigger A means ERM takes longer to recover. For $|A| = 500$, still $> 10\%$ error at $n = 500$. This confirms: as A grows, ERM needs proportionally more data.

6.3 Experiment 3: Local Regularization Strategies



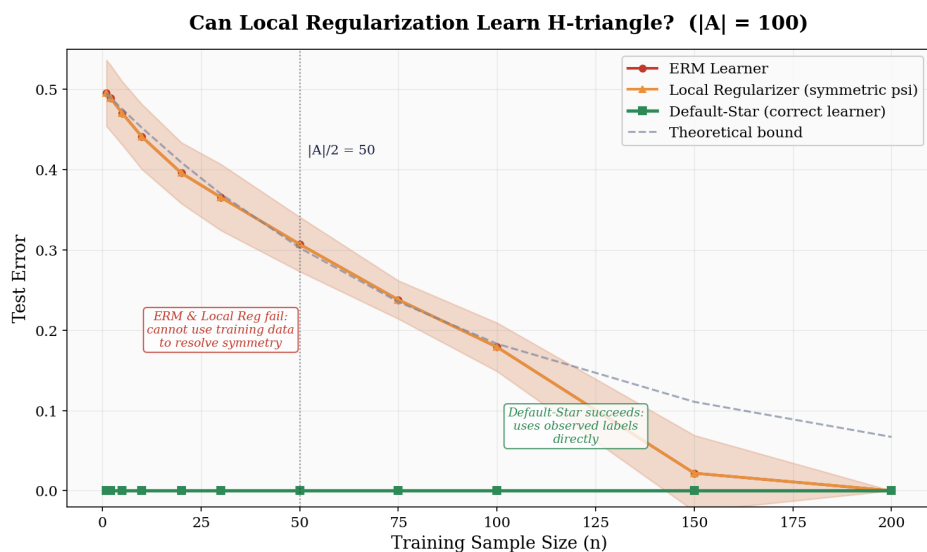
I tested three ψ strategies. Random/symmetric (orange) performs the same as ERM. Prefer-star and prefer-small-sets (blue, purple) are even *worse*. None come close to Default-Star (green, flat at zero). No ψ I tried could beat the symmetry.

6.4 Experiment 4: Symmetry Visualization



Left: set structure with seen/unseen markers. Right: the argument — for an unseen point, both the correct and adversarial hypotheses have zero training error, and $\psi(h, x)$ gets the exact same information for both. No way to tell them apart.

6.5 Experiment 5: Summary



ERM (red) and local regularizer (orange) overlap almost perfectly. The theoretical bound matches. Default-Star (green) is flat at zero. The takeaway: ERM and local reg fail identically because neither can use training data to break the symmetry.

7 What This All Means

If local regularization turns out to be **sufficient**: we'd get a simple learning template for multiclass (like ERM for binary), and it would mean the unlabeled data in OIG learners is unnecessary.

If it turns out to be **insufficient**: the OIG approach is vindicated, training data geometry really matters, and there’s a real limit on what sample-independent methods can do.

My experiments point toward “insufficient” — every ψ I tested failed on $\mathcal{H}_\Delta$. But this is experimental evidence, not a formal proof. The gap is also narrow: unsupervised local regularization (which CAN see unlabeled data) already works for everything.

8 Conclusion

1. Multiclass is harder than binary — ERM doesn’t cut it, you need improper learners
2. Local regularization ($\psi(h, x)$) is the best candidate for a simple multiclass template
3. $\mathcal{H}_\Delta$ is a clever counterexample exploiting the fact that ψ can’t see training data
4. My experiments confirm: ERM and all tested local regularizers fail on $\mathcal{H}_\Delta$, Default-Star succeeds
5. The open problem is still open, but evidence leans toward “no”

References

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